- Mesophiles
- Thermophiles
- Hyperthermophiles

Psychrophiles

A psychrophile can be defined as an organism with an optimal growth temperature of 15°C, maximum growth temperature of 20°C and a minimum growth temperature at 0°C. These organisms are found in polar regions like Arctic and Antarctic oceans. They are rapidly killed as the temperature rises because the cellular constituents start to leak due to cell membrane disruption. Some examples of psychrophiles are *Moritella, Photobacterium* and *Pseudomonas.*



Snow alga – *Chlamydomonas nivalis* grows within the snow and its brilliant red coloured spores are responsible for the formation of pink snow.



HOTS

Why do unopened pasteurized milk spoil even under refrigeration?

Psychrotolerant

Organisms that can grow at 0°C, but have temperature optimum growth temperature range of 20°C-40°C are called psychrotolerant.

Mesophiles

These are microorganisms that grow in optimum temperature between 20-45°C, they have a temperature minimum of 15-20°C and a maximum temperature of 45°C. All human pathogens are mesophiles.

Thermophiles

Organisms whose growth temperature optimum is between 55-65°C are called thermophiles. They have minimum growth temperature of 45°C. These organisms are found in compost stacks, hot water lines and hot springs. They contain enzymes that are heat stable and protein synthesis systems function well at high temperature.

Infobits

Taq polymerase, a DNA polymerase enzyme which is of great applied importance used in DNA amplification. It is isolated from *Thermus aquaticus*, a thermophile.

Hyperthermophiles

Organisms whose growth optimum temperature is above 80°C are called hyperthermophiles. These are mostly bacteria and archaebacteria. They are found in boiling hot springs and hydrothermal vents on seafloor.

2. pH

pH is defined as the negative logarithm of the hydrogen ion concentration. pH scale extends from pH 0.0 to pH 14.0 and each exchange of 1 pH unit represents a 10 fold change in hydrogen ion concentration. pH greatly influences microbial growth. Each organism has a definite pH range and well defined pH growth optimum. Most natural environments have pH values between 5 and 9.

TN_GOVT_XI_Micro_Biology_CH06.indd 79 16-02-2019 9.58.11 AM